

Topic to discuss

→ Proof of Newton Divided difference.

YouTube @StartPracticing

Newton divided difference formula

$$y = f(x) = y_0 + (x-x_0) \Delta y_0 + (x-x_0)(x-x_1) \Delta^2 y_0 + \\ (x-x_0)(x-x_1)(x-x_2) \Delta^3 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)(x-x_3) \Delta^4 y_0 + \dots$$

OR

$$y = f(x) = y_0 + (x-x_0) [x_0, x_1] + (x-x_0)(x-x_1) [x_0, x_1, x_2] \\ + (x-x_0)(x-x_1)(x-x_2) [x_0, x_1, x_2, x_3] + \\ (x-x_0)(x-x_1)(x-x_2)(x-x_3) [x_0, x_1, x_2, x_3, x_4] + \\ \dots$$

Proof:

Let $y_0, y_1, \dots, y_n$ be the values of $y=f(x)$ corresponding to the arguments $x_0, x_1, x_2, \dots, x_n$.

From the definition of divided difference,

we have,

$$[x, x_0] = \frac{y_0 - y}{x_0 - x} = \frac{y - y_0}{x - x_0}$$

$$\Rightarrow y = y_0 + (x - x_0)[x, x_0] \quad \text{--- (1)}$$

$$\text{Again, } [x, x_0, x_1] = \frac{[x_0, x_1] - [x, x_0]}{x_1 - x} = \frac{[x, x_0] - [x_0, x_1]}{x - x_1}$$

$$\Rightarrow [x, x_0] = [x_0, x_1] + (x - x_1)[x, x_0, x_1] \quad \text{--- (2)}$$

Lets substitute equation —(2) in equation —(1)
we get ,

$$y = y_0 + (x - x_0) [x, x_0]$$

$$\Rightarrow y = y_0 + (x - x_0) \left[[x_0, x_1] + (x - x_1) [x, x_0, x_1] \right]$$
$$\Rightarrow y = y_0 + (x - x_0) [x_0, x_1] + (x - x_0)(x - x_1) [x, x_0, x_1] \quad \text{--- (3)}$$

similarly,

$$[x, x_0, x_1, x_2] = \frac{[x, x_0, x_1] - [x_0, x_1, x_2]}{x - x_2}$$

$$[x, x_0, x_1] = [x_0, x_1, x_2] + (x - x_2) [x, x_0, x_1, x_2] \quad \text{--- (4)}$$

Substitute equation —(4) in equation —(3)

$$\Rightarrow y = y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x, x_0, x_1] \quad \text{--- (3)}$$

$$\Rightarrow y = y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)\left[[x_0, x_1, x_2] + (x-x_2)[x, x_0, x_1, x_2]\right]$$

$$\Rightarrow y = y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] + (x-x_0)(x-x_1)(x-x_2)[x, x_0, x_1, x_2]$$

Proceeding this manner, we get,

$$y = f(x) = y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] \\ + (x-x_0)(x-x_1)(x-x_2)[x_0, x_1, x_2, x_3] \\ + \dots$$

which can also be written as,

$$y = f(x) = y_0 + (x-x_0) \Delta y_0 + (x-x_0)(x-x_1) \Delta^2 y_0 + \\ (x-x_0)(x-x_1)(x-x_2) \Delta^3 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)(x-x_3) \Delta^4 y_0 + \dots$$

(Proved)
=.

Connect with me



[Start Practicing](#)



[i._am._arfin](#)



[Arfin Parween](#)



[Arfin Parween \(.Subscribe here too\)](#)



[Arfin Parween](#)



[Arfin Parween](#)

YouTube @StartPracticing